# Part 4: Methods and Results

## Scope and research questions

We study PJM nodal day-ahead LMP (primary price signal) with NOAA Integrated Surface Database (ISD) weather aligned to each pricing node. **RQ1** asks whether the strength of the weather–grid *association* (correlation) has shifted or differs between nodes classified as data-center (DC)–likely versus not. **RQ2** asks whether *weather-only* predictive models exhibit systematically different error in DC-likely versus other nodes. All tables and figures below are derived from analysis exports or model metrics JSON—not ad hoc screenshots of spreadsheets.

## Data acquisition and alignment

**Energy.** Day-ahead full-system LMP is read from PJM parquet archives (hourly `total_lmp_da` by `pnode_id` and timestamp); load-related rows use forecast load fields where included in the training script. Timestamps are parsed and aligned to UTC.

**Weather.** For each node, ISD hourly CSVs are read by station and year; fields are converted to physical units (temperature and dew point from tenths °C, sea-level pressure from tenths hPa, wind from tenths $\mathrm{m\,s^{-1}}$, precipitation from tenths mm). Sentinel codes are treated as missing. Cooling and heating degree hours (CDH, HDH) use base 18 °C; relative humidity uses a Magnus-form estimate from temperature and dew point.

**Merge.** For each node–year, energy and weather series are merged on time using `pandas.merge_asof` in UTC with a 30 minute tolerance so each LMP hour matches the nearest weather observation; unmatched hours are excluded from correlation and feature matrices.

## Data-center labeling (for bucket comparisons)

Nodes are scored in `scripts/export_pjm_metadata.py` using `scripts/dc_region_scoring.py` (geographic proximity to known DC-heavy footprints, optional attributes, confidence). Each node receives `classificationLabel` and `isDataCenterHeavy`; correlation and model rows carry `dc` vs. `nonDc` buckets consistent with that export. Reproducibility requires re-running the metadata export before rebuilding correlations or models when labels change.

## RQ1: correlation construction

For each node, calendar year, weather variable (e.g. CDH, HDH, temperature, RH, wind, pressure, precipitation), and target (`lmp` and/or `load`), we compute Pearson correlation between aligned weather and target series. Period tags follow the export script: `earlyAI` corresponds to 2021 rows and `recentAI` to 2022–2025. The app-facing file `data/exports/correlations_by_region_period.json` stores each coefficient with `nodeId`, `regionId`, `year`, `isDataCenterHeavyBucket`, and `target`.

## RQ2: predictive models and metrics

**Features.** Weather-based inputs (e.g. `temp_c`, `dew_c`, `rh_pct`, `wind_ms`, `slp_hpa`, `precip_mm`, `cdh`, `hdh`) plus simple calendar indicators as implemented in `scripts/train_rq2_models.py`.

**Models.** Six scikit-learn pipelines: LinearRegression, Ridge ($\alpha$=1), Lasso ($\alpha$=0.001, high iteration cap), RandomForest (90 trees), GradientBoosting, ExtraTrees (100 trees). Continuous inputs use median imputation; linear pipelines add `StandardScaler`.

**Evaluation.** Chronological train/test split with holdout starting at `--test-start-year` (default 2024) so test metrics reflect later calendar years. We record RMSE, MAE, $R^2$, and weather-only nRMSE = RMSE/$|\bar{y}|$ on the test split when $\bar{y}$ is available. Rows are tagged by node, period, target,

model name, and DC bucket. Output: `data/exports/model_performance.json`.

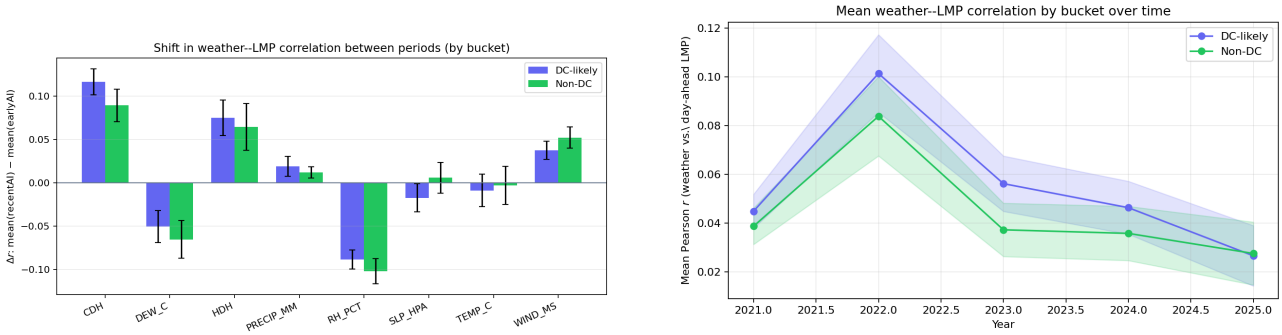
### Between-group statistical summaries

`scripts/build_hypothesis_exports.py` reads `nodes.json`, correlations, and model metrics. For RQ1 it aggregates node-level mean absolute correlation per DC bucket; for RQ2 it uses weather-only rows and prefers $n$RMSE when present. It computes bootstrap 95% CIs (5000 resamples, fixed seed) for DC minus non-DC mean differences when group sizes exceed configured gates, and two-sided permutation $p$-values (3000 shuffles) when eligible; otherwise rows are marked descriptive-only.

### Figure generation (for the report)

Trend panels (Fig. 1) are produced by `docs/generate_report_figures.py` from the correlation export (LMP target). Distribution and mean-difference panels (Figs. 2–3) are produced by `docs/generate_rq_research_figures.py`, which reads the same exports plus `hypothesis_tests.json`. After changing data, run those scripts before rebuilding this PDF.

### Research question 1

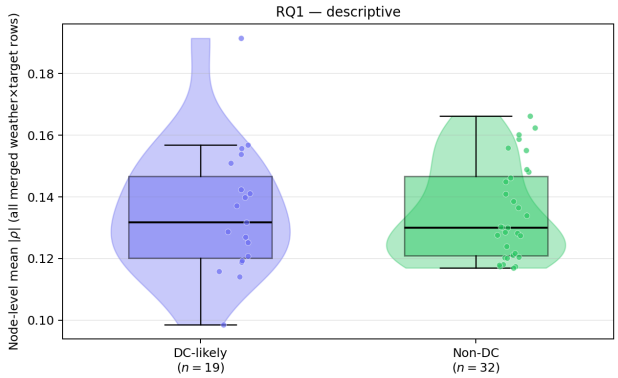



**(a)** Mean change in weather–LMP $r$ from `earlyAI` to `recentAI` by bucket and variable.

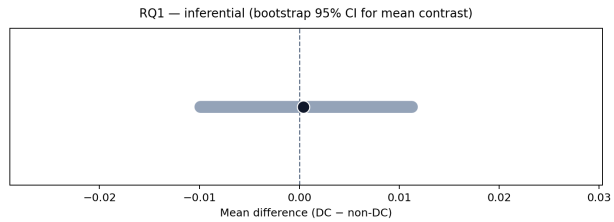
**(b)** Mean $r$ by calendar year and bucket (±1 SEM).

**Figure 1:** RQ1: how association moves across periods and years (DC-likely vs. non-DC).

**Description and relevance.** Figure 1 addresses RQ1 directly: panel (a) shows whether weather–price linear association strengthened or weakened between our early (2021) and recent (2022–2025) windows for each weather variable, separately for DC-likely and non-DC nodes; panel (b) shows whether the two buckets trace similar yearly paths or diverge.




**(a)** Distribution of node-level mean $|\rho|$ by bucket ($n$ on axis).




**(b)** Bootstrap 95% CI for DC − non-DC mean of node-level mean $|\rho|$.

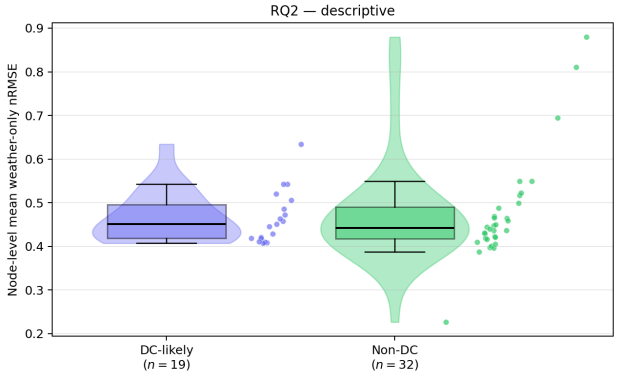
**Figure 2:** RQ1: cross-sectional spread and uncertainty for the bucket mean contrast.

**Description and relevance.** Figure 2 summarizes whether typical association *magnitude* differs between buckets when each node is collapsed to one mean absolute correlation (panel (a)), and whether the difference of *group means* is pinned away from zero at our sample scale (panel (b)); Table 1 gives the numeric CI and $p$-value for that contrast.

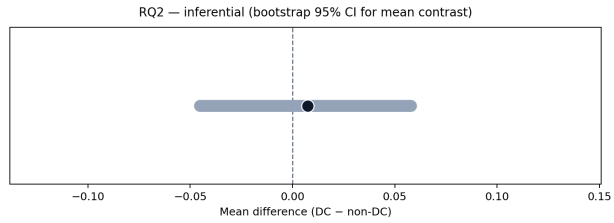
### Significance for RQ1

Taken together, Figs. 1–2 and Table 1 suggest that **DC-likely PJM nodes are not clearly separated from other nodes by a stronger or weaker weather–price association in the aggregate**: yearly tracks overlap, within-bucket spread is large, and the formal DC minus non-DC contrast on mean node-level mean $|r|$ is negligible in magnitude (means both $\approx 0.135$), with a 95% CI spanning zero ($[-0.010, 0.011]$) and $p = 0.945$. Interpreting RQ1 in substantive terms, we do *not* see evidence here that "data-center-heavy context" coincides with a distinct *population mean* correlation profile relative to non-DC nodes under our labeling and time split; weather remains a measurable correlate for both groups, and time variation appears shared rather than uniquely divergent for DC-likely nodes.

## Research question 2




**(a)** Per-node mean weather-only $n$RMSE (or RMSE) by bucket.




**(b)** Bootstrap 95% CI for DC − non-DC mean $n$RMSE on evaluation rows.

**Figure 3:** RQ2: weather-only error distributions and mean contrast.

**Description and relevance.** Figure 3 targets RQ2: panel (a) shows whether prediction error concentrates differently across DC-likely vs. other nodes when each node is summarized by its mean

weather-only error; panel (b) matches the hypothesis pipeline's *row-level* mean contrast for $n$RMSE (multiple evaluation rows per node possible).

### Significance for RQ2

Figure 3 and Table 1 suggest that **DC-likely nodes are not convincingly "harder to predict" with weather-only models at the level of the population mean error in this run**: distributions overlap heavily, and the DC minus non-DC mean $n$RMSE contrast is small ($\approx 0.465$ vs. $\approx 0.457$, $\approx 1.6\,\%$ relative) with a CI that includes zero ($[-0.045, 0.058]$) and $p = 0.791$. Substantively, we should not conclude that data-center-adjacent pricing nodes are systematically less weather-explainable by this metric; any slight descriptive gap is dominated by node-to-node and model-configuration variability.

**Table 1:** Formal bucket contrasts from `hypothesis_tests.json` (bootstrap CI for DC − non-DC mean). Not a spreadsheet screenshot—exported statistical summary.

| RQ | Metric | $n$ DC/ non | Means (DC; non) | 95% CI; $p$; Cohen's $d$ |
|---|---|---|---|---|
| 1 | Mean $\lvert r\rvert$/node | 19/32 | 0.135; 0.135 | $[-0.010, 0.011]$; 0.945; 0.022 |
| 2 | $n$RMSE (rows) | 324/270 | 0.465; 0.457 | $[-0.045, 0.058]$; 0.791; 0.023 |

## Reproducibility

- Regenerate trend and RQ figures: `pythondocs/generate_report_figures.py`; `pythondocs/generate_rq_research_figures.py`.
- Refresh metrics and hypothesis JSON: `pythonscripts/train_rq2_models.py` (configure years, node limits, test year); `pythonscripts/build_hypothesis_exports.py`; `npmrunsync:data`.
- Repository: `https://github.com/jaiparmar940/energy-weather-app`; full data layout and Box bundle instructions are in `README.md`.